

SKILL- Week 1

```
harshitha18@LAPTOP-6E22AR X + v
harshitha18@LAPTOP-6E22AR6C:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [447 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [111 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [84.7 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [235 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [76.9 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [166 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [331 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [63.0 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.4 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3032 B]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [371 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [89.6 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [32.8 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [321 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [61.2 kB]
Fetched 2723 kB in 7s (413 kB/s)
40 packages can be upgraded. Run 'apt list --upgradable' to see them.
harshitha18@LAPTOP-6E22AR6C:~$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 40
harshitha18@LAPTOP-6E22AR6C:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
GNU nano 7.2 hello.c
#include <stdio.h>

int main()
{
    printf("Hello, World!\n");
    return 0;
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        [ Read 8 lines  ^G Location   M-U Undo      M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify      ^_ Go To Line M-E Redo      M-G Copy       ^Q Where Was
```

```
harshitha18@LAPTOP-6E22AR6C:~$ nano hello.c
harshitha18@LAPTOP-6E22AR6C:~$ gcc hello.c -o hello
harshitha18@LAPTOP-6E22AR6C:~$ ./hello
Hello, World!
harshitha18@LAPTOP-6E22AR6C:~$ |
```

The Linux environment was successfully configured with GCC, and the required project structure was created using basic Linux commands.

The `fork()` and `exec()` system calls were also understood and implemented to demonstrate process creation and command execution.